CartCycle: An Explainable End-to-End Analytics Pipeline for Customer Segmentation, Churn Prediction and Demand Forecasting in Online Retail

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***Abstract*—Online retailers generate large volumes of transactional data, yet small and mid-sized businesses still analyse customers, sales and inventory in separate tools, which delays decisions and hides the link between acquisition, retention and revenue. This paper presents CartCycle, a full-cycle e-commerce analytics pipeline that unifies customer analytics and sales analytics. The pipeline covers data validation and cleaning, recency–frequency–monetary (RFM) feature engineering, K-means segmentation, gradient-boosted churn prediction with feature-importance explanations, market-basket mining, time-series forecasting and a no-code dashboard that converts model output into retention, bundling and restocking actions. The pipeline and dashboard are run on the public CDNOW transaction dataset of 23,502 customers. K-means with $k = 3$ gives a silhouette score of 0.480, and the churn classifier reaches an accuracy of 0.818, an F1-score of 0.888 and an AUC-ROC of 0.803 on a held-out set. A damped-trend forecaster does not improve on a naive baseline for this single-cohort series, which shows why every forecast is reported against a baseline. The paper also discusses privacy, deployment and limitations, and outlines future work on streaming, recommendation and privacy-preserving analytics.**

***Index Terms*—E-commerce analytics, customer segmentation, churn prediction, RFM analysis, demand forecasting, business intelligence**

I. Introduction

Electronic commerce has grown into one of the largest channels of retail, and every click, basket and payment now leaves a digital trace. For a retailer, these traces are an asset only when they are converted into decisions about whom to target, what to stock and when to promote. In practice, many small and mid-sized online businesses keep customer records, order histories and product catalogues in separate spreadsheets, storefront exports and reporting tools. Analysts spend most of their time reconciling these sources by hand, reports arrive late, and the relationship between acquiring a customer, retaining that customer and the revenue that follows remains difficult to see.

CartCycle is proposed as a full-cycle answer to this problem: one pipeline that starts from raw transactions and ends in an interactive dashboard that recommends actions, covering both the customer view and the sales view of the business. The main contributions of this paper are as follows.

- A modular, layered design that shares one cleaned data source between customer analytics and sales analytics.
- An explainable customer branch that combines RFM segmentation, lifetime value ranking and churn prediction.
- A sales branch that combines revenue tracking, basket mining and baseline-checked forecasting.
- An evaluation of the segmentation, churn and forecasting modules on a public retail transaction dataset, with an honest account of where the approach does and does not help.

II. Related Work

The foundations of customer analytics in retail are well established. Recency, frequency and monetary value have long been used to describe customer behaviour, and probabilistic models of repeat buying estimate how many customers remain active and what they are likely to spend next [1]–[3]. Clustering methods such as K-means group customers into behavioural segments without labelled data [4], [5], and the silhouette coefficient offers a way to judge how well separated those groups are [6]. Case studies on public online retail transactions show that segmentation built on these features can guide targeted marketing [7].

For prediction, ensemble methods such as random forests and gradient-boosted trees perform strongly on tabular customer data [8]–[10], and model-agnostic explanation techniques allow their outputs to be inspected by non-specialists [11]. On the sales side, association rule mining reveals products that are bought together [12], [13], while classical and decomposable time-series models capture trend and seasonality for demand forecasting [14], [15]. Recommender system research adds methods for personalising what each shopper sees [16].

Most of these contributions, however, are studied in isolation, and each is usually presented as a single algorithm rather than as part of a working system that a retailer can operate every day.

III. Motivation

The justification for this work follows from that gap. First, the separation of customer analytics from sales analytics

TABLE I
FEATURES USED IN CARTCYCLE

Feature	Code	Purpose
Recency (days)	REC	Time since last purchase
Frequency	FREQ	Number of distinct purchase dates
Monetary value	MON	Total spend of a customer
Average order value	AOV	Typical basket size
Quantity	QTY	Total items bought
Tenure (days)	TENURE	Time since first purchase
Product category	CAT	Category-level sales analysis
Churn flag	CHURN	Target label for prediction

forces business users to switch tools and to trust numbers that were computed on different cleaning rules, so a single governed pipeline improves consistency and saves analyst time. Second, acquiring a new customer is generally more costly than retaining an existing one, which makes early identification of at-risk customers economically important, particularly for retailers whose margins are thin. Third, poorly forecast demand leads either to stock-outs that lose sales or to excess inventory that ties up capital. Fourth, commercial analytics suites are often priced and configured for large enterprises, and an open-source-based platform lowers the barrier for smaller sellers. Finally, predictions that cannot be explained are rarely acted upon, so CartCycle treats interpretability and an explicit action layer as design requirements.

IV. PROPOSED METHODOLOGY

A. Data Preparation and Features

The data consist of three linked tables: orders with line items, customers, and products. After loading, the pipeline validates data types and ranges, removes exact duplicate rows, separates cancelled and returned orders from completed sales, resolves missing or inconsistent identifiers, and standardises dates and currency fields. Unit price and quantity outliers are flagged rather than deleted, so that bulk purchases are not mistaken for errors. The feature stage derives the attributes summarised in Table I.

The overall workflow is shown in Fig. 1. Cleaned data flow into two parallel analytical branches that share the same feature store, and their outputs are evaluated before being published to the dashboard. This keeps every figure on the dashboard traceable to a single cleaned source and allows each module to be retrained or replaced without disturbing the others.

B. System Architecture

CartCycle is organised as the layered architecture shown in Fig. 2. Source files enter an ingestion layer that applies the cleaning rules, and the cleaned tables and features are written to a storage layer that acts as the single source of truth. The analytics engine reads from this layer to train and score the models, and a service layer schedules retraining, keeps versioned copies of fitted models and prepares exports. The presentation layer only reads finished results, so a slow model run never blocks a user browsing the dashboard. The

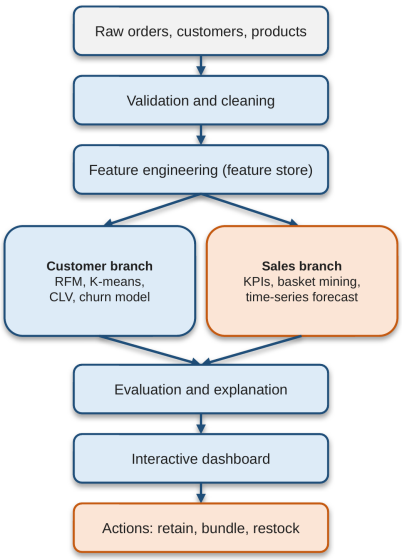


Fig. 1. Workflow of the proposed CartCycle pipeline.

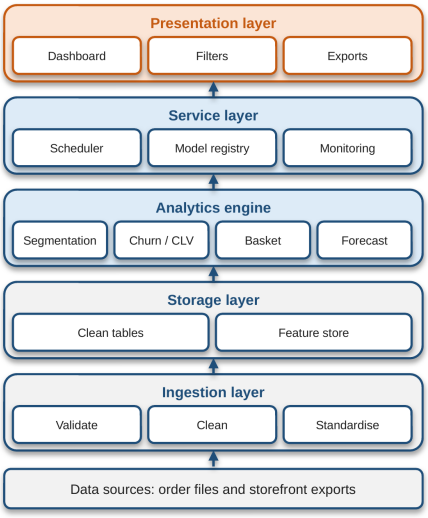


Fig. 2. Layered architecture of CartCycle.

pipeline is written in Python using widely available libraries [17], [18], which keeps software cost low and allows the code to be audited by a small team.

C. Customer Analytics Branch

Each customer is described by the days since the last purchase, the number of distinct purchase dates, and the total and average amount spent. The log-transformed RFM features are standardised and clustered with K-means, and the number of clusters is chosen by combining the elbow curve with the silhouette score so that segments are compact and meaningful to marketers. Each segment is given a plain-language label based on its centroid. Customer lifetime value is ranked from historical spend, frequency and tenure.

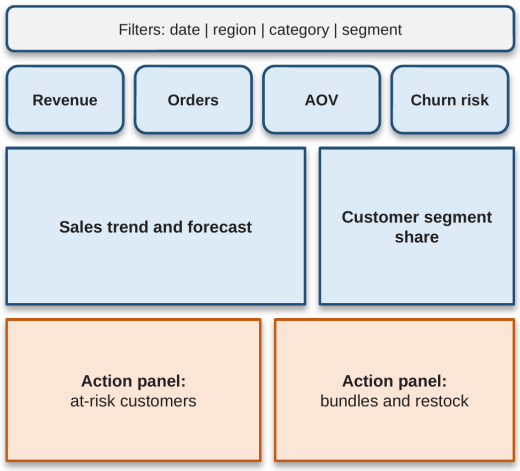


Fig. 3. Wireframe of the CartCycle dashboard (layout illustration).

Churn is formulated as binary classification: a customer is considered lost when no purchase occurs within a defined inactivity window after a cut-off date. A gradient-boosted tree classifier [10] is trained on features computed strictly before the cut-off to prevent information leakage. Feature importance shows which behaviours, for example a long gap since the last order, drive predictions; Shapley-value explanations [11] are the planned extension.

D. Sales Analytics Branch

Completed orders are aggregated by day, week and month to produce revenue, order count, units sold and average order value, and these are broken down by category, product and region. Market-basket analysis applies the Apriori algorithm to order baskets and reports support, confidence and lift so that only associations that are both frequent and stronger than chance are surfaced as bundle suggestions. For demand planning, weekly revenue is forecast with damped-trend exponential smoothing and always compared against naive and moving-average baselines, so that any real gain in accuracy is visible.

E. Dashboard and Action Layer

The dashboard is the point at which analysis becomes action. Users filter by date range, region, category and segment, and every chart updates from the same cleaned data. An action layer translates model output into recommendations: customers with a high churn probability and high lifetime value are listed for retention offers, frequently co-purchased products are proposed as bundles, and categories with rising forecast demand are flagged for restocking. The layout of the main page is sketched in Fig. 3. It follows the order in which a manager normally reasons, moving from a headline number to its cause and then to a decision.

V. Experimental Setup

A. Dataset

The pipeline was evaluated on the public CDNOW transaction sample, an online music retailer dataset widely used in customer-base analysis [2], [19]. After removing duplicate and non-positive records, it contains 69,324 transactions from 23,502 customers between 1 January 1997 and 30 June 1998, with a total revenue of \$2,495,982.75. The dataset has no product category or basket information, so the basket-mining module could not be evaluated; all other modules, including the dashboard, were run on it.

TABLE II
Segment Profiles ($k = 3$, silhouette = 0.480)

Segment	Share (%)	REC	FREQ	MON (\$)	Churn
Loyal	13.9	39.7	7.00	298.20	0.33
At risk	28.0	191.9	2.68	103.55	0.70
Dormant	58.1	316.6	1.08	26.73	0.91

B. Protocol and Metrics

Features were computed from transactions up to the cut-off date of 31 December 1997. A customer is labelled as churned if no purchase occurs between 1 January and 30 June 1998, which gives a churn rate of 77.1%. The customers were split into 70% training and 30% test sets with stratification. The classifier is evaluated with accuracy, precision, recall, F1-score and AUC-ROC, segmentation with the silhouette score, and forecasting with mean absolute percentage error (MAPE) on the 26 weeks after the 52-week training period. Because churners are the majority class here, a classifier that always predicts churn would reach 0.771 accuracy, so the other metrics are essential.

VI. Results and Discussion

A. Customer Segmentation

The elbow curve and silhouette scores for $k = 3$ to 7 are shown in Fig. 4. The silhouette score is highest at $k = 3$ (0.480), so three segments were used. The segment profiles are given in Table II and visualised in Fig. 5. A small loyal segment (13.9% of customers) buys about seven times with high spend and has a churn rate of 33%, whereas the dormant segment (58.1%) bought about once, long ago, and 91% of it churned. The middle segment is a natural target for retention campaigns.

B. Churn Prediction

The churn classifier results on the held-out test set are given in Table III, and the ROC curve and feature importances in Fig. 6. The AUC-ROC of 0.803 shows useful ranking ability, and the recall of 0.938 means most churners are caught, at the cost of some false alarms (precision 0.844). Recency and frequency-related features are the strongest drivers, which agrees with the customer-base literature [1].

C. Demand Forecasting

Table IV compares the forecasters, and Fig. 7 plots them against the actual weekly revenue. The damped-trend model (MAPE 27.12%) is no better than the naive baseline (27.12%), and both are clearly better than an eight-week moving average

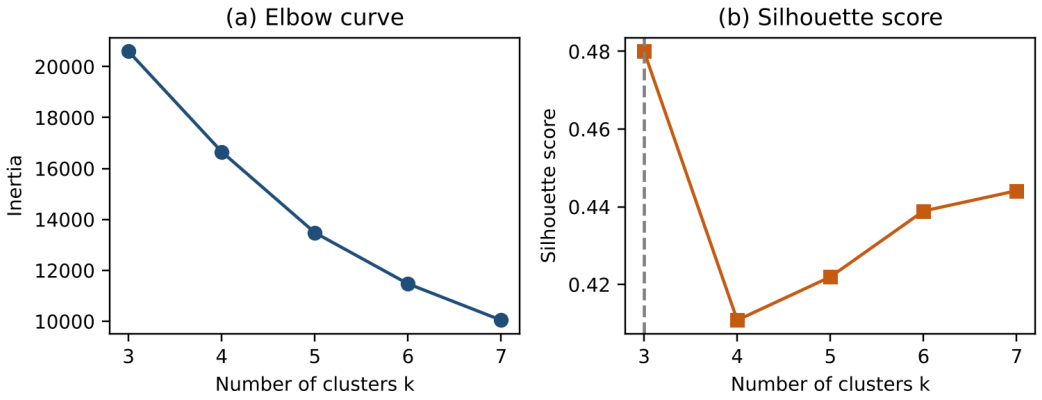


Fig. 4. Choosing the number of clusters: (a) elbow curve and (b) silhouette score.

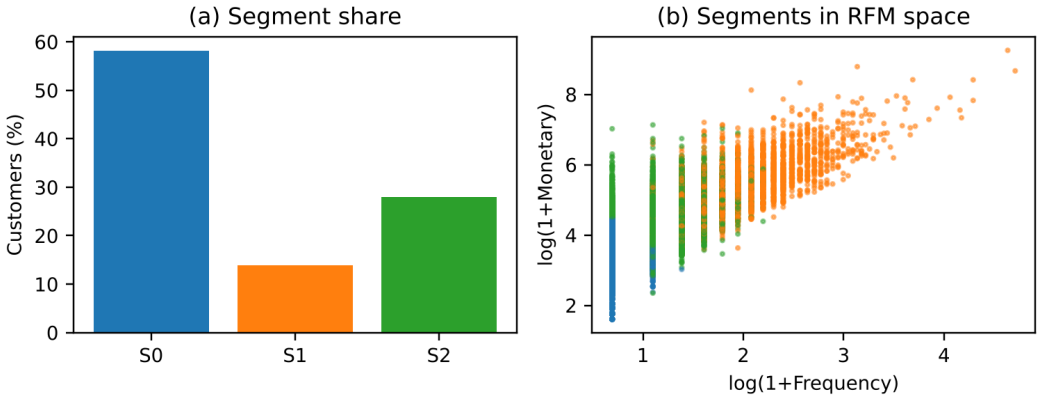


Fig. 5. Customer segments: (a) share of customers per segment and (b) segments in RFM space.

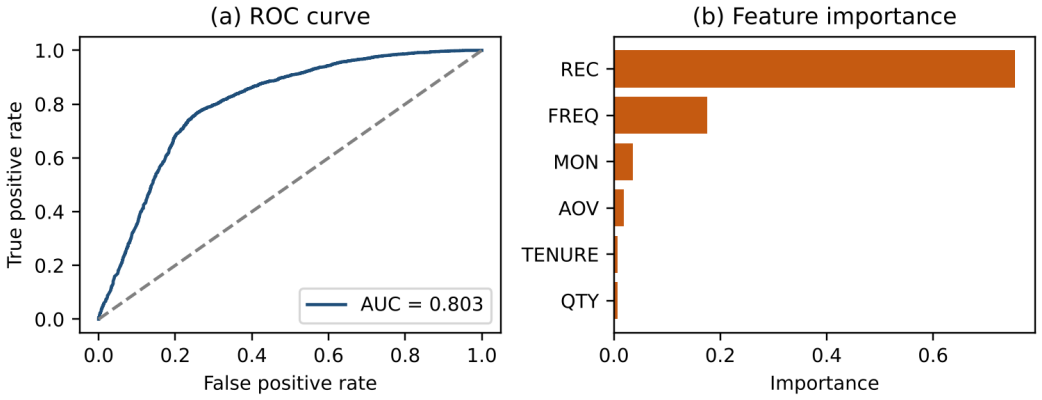


Fig. 6. Churn model output: (a) ROC curve and (b) feature importance.

(41.34%). This is an honest negative result: the series is dominated by the decay of a single acquisition cohort, and 52 weeks of training data cannot support seasonal terms. It shows why CartCycle always reports forecasts next to a baseline, and that gains from seasonal or decomposable models [14], [15] should be expected only on multi-year, multi-cohort data.

D. Dashboard Output

The implemented dashboard was run on the CDNOW data, and its main page is shown in Fig. 8. The indicator row reports revenue of $2,495,983, 67,511 orders (distinct customer purchase dates), an average order value of $36.97, 23,502 customers and an average churn risk of 77.1%, which agrees

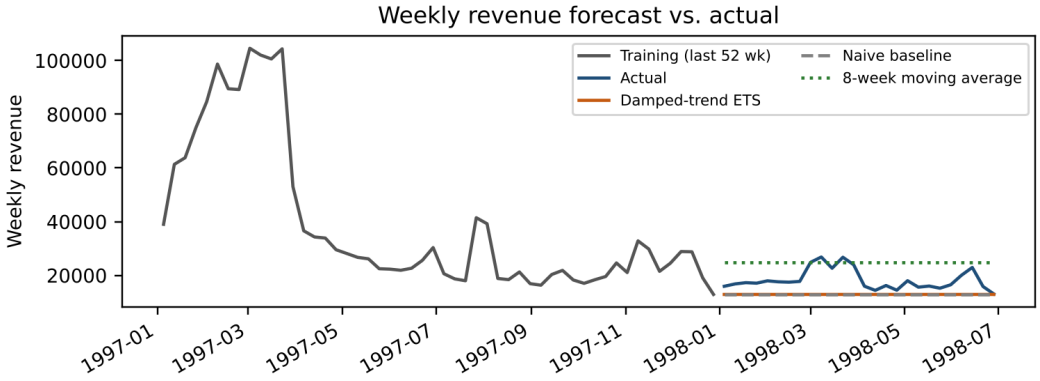


Fig. 7. Weekly revenue forecast compared with actual values and baselines.

TABLE III
EVALUATION OF THE CHURN CLASSIFIER ON THE TEST SET

Metric	Value
Accuracy	0.818
Precision	0.844
Recall	0.938
F1-score	0.888
AUC-ROC	0.803

TABLE IV
FORECAST ACCURACY OVER 26 TEST WEEKS

Method	MAPE (%)
Naive (last value)	27.12
Damped-trend exponential smoothing	27.12
8-week moving average	41.34

with the churn rate in Section V-B. The weekly revenue trend shows the sharp decay after the first-quarter 1997 acquisition cohort, and the segment chart confirms the 58.1%, 28.0% and 13.9% split of Table II. The filter panel on the left restricts every panel by date range and customer segment.

Fig. 9 shows the action layer. The priority retention list ranks customers whose churn probability is high by the value at risk, which is the churn probability multiplied by the customer's total spend, and it can be exported as a CSV file for campaign teams. The restocking panel compares the forecast with recent demand and, for this data, raises no flag. Because CDNOW contains no product column, the dashboard reports that bundle suggestions are unavailable instead of showing empty output.

E. Practical Considerations

Integrated use has several advantages over separate tools. Cleaning is performed once and shared, so revenue totals and customer counts agree across every view, and segment labels and churn scores appear next to sales trends, which lets an analyst see whether a revenue dip comes from fewer active customers or from smaller baskets.

Responsible handling of customer data is part of the design. Personal identifiers are replaced with pseudonymous keys before features are created, only the fields required for a stated purpose are retained, and dashboard outputs are aggregated unless a user has permission to view individual lists. Segment labels and risk scores are reviewed for unintended bias. For deployment, customer features and forecasts are recomputed on a schedule, the performance of each deployed model is logged, and when error rises above an agreed tolerance the model is retrained and the previous version remains available for rollback.

F. Limitations

Several limitations should be acknowledged. The evaluation uses one public dataset from the late 1990s with no product information, so basket mining and lifetime value still need validation on a modern multi-category store, where the dashboard's bundle suggestions can also be exercised. Output quality depends on the completeness of the source data, and new customers have too little history for reliable estimates, which is the cold-start problem. Churn definitions depend on the inactivity window. Observational data cannot prove that an offer caused a customer to stay, so uplift should be confirmed through controlled experiments [20].

VII. CONCLUSION AND FUTURE WORK

CartCycle shows that customer analytics and sales analytics can be combined into a single, explainable and affordable pipeline that takes a retailer from raw transactions to concrete actions. On the CDNOW data, the pipeline and dashboard run end to end, the segmentation and churn modules give a silhouette score of 0.480 and an AUC-ROC of 0.803, while the forecasting experiment shows the need for baseline checks. Planned extensions include a streaming architecture for near real-time scoring, a personalised recommendation engine [16], sequence and survival models, A/B testing modules [20], and privacy-preserving techniques such as federated learning with scheduled drift monitoring.

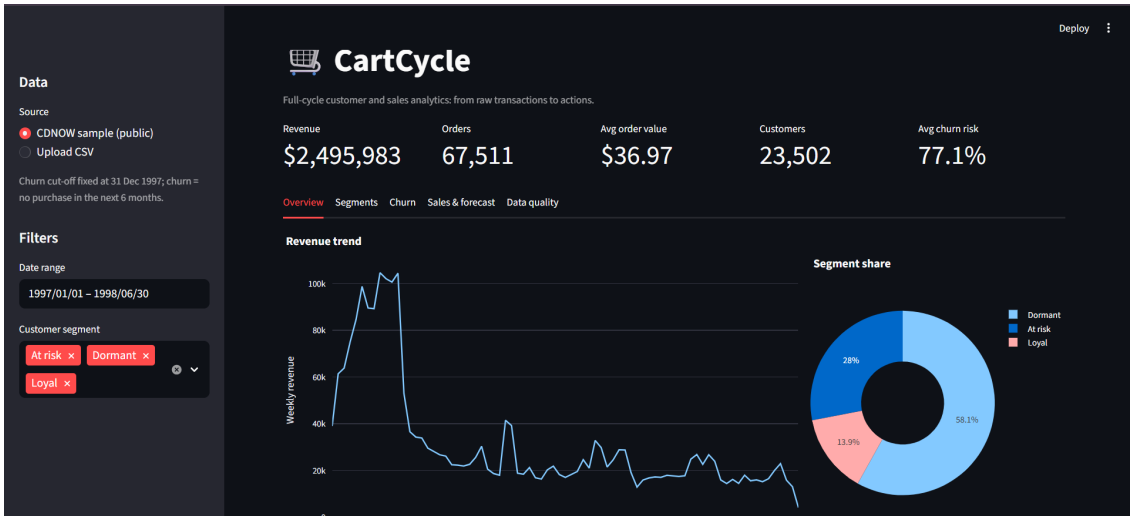


Fig. 8. Main page of the CartCycle dashboard on the CDNOW data: indicators, revenue trend and segment share.

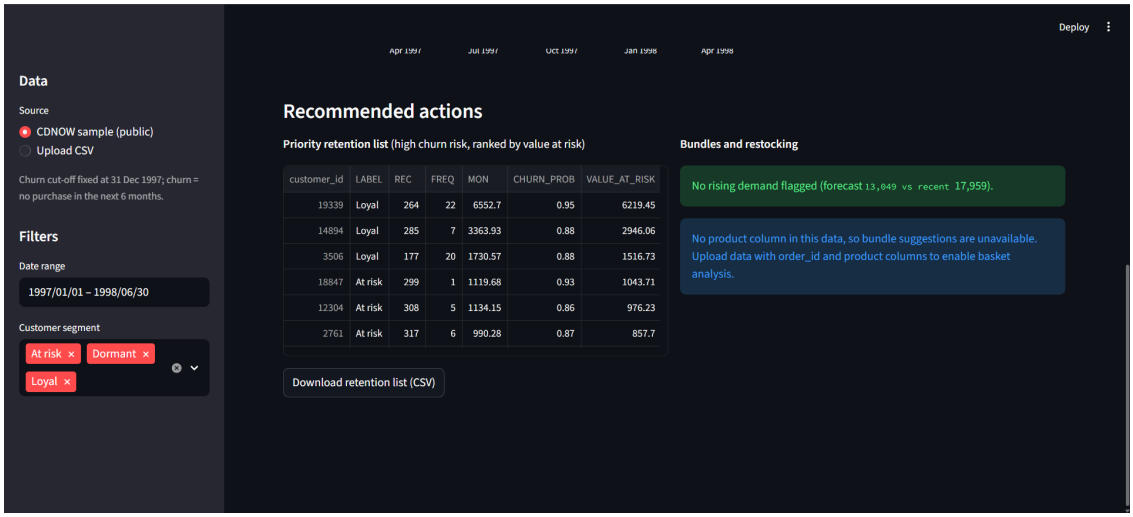


Fig. 9. Action layer of the dashboard: priority retention list with CSV export and the restocking and bundle panel.

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